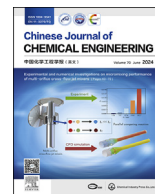




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Full Length Article

CFD-PBM coupled modeling of the liquid–liquid dispersion characteristics and structure optimization for Kenics static mixer

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ABSTRACT

Kenics static mixers (KSM) are extensively used in industrial mixing–reaction processes by virtue of high mixing efficiency, low power homogenization and easy continuous production. Resolving liquid droplet size and its distribution and thus revealing the dispersion characteristics are of great significance for structural optimization and process intensification in the KSM. In this work, a computational fluid dynamics–population balance model (CFD-PBM) coupled method is employed to systematically investigate the effects of operating conditions and structural parameters of KSM on droplet size and its distribution, to further reveal the liquid–liquid dispersion characteristics. Results indicate that higher Reynolds numbers or higher dispersed phase volume fractions increase energy dissipation, reducing Sauter mean diameter (SMD) of dispersed phase droplets and with a shift in droplet size distribution (DSD) towards smaller size. Smaller aspect ratios, greater blade twist and assembly angles amplify shear rate, leading to smaller droplet size and a narrower DSD in the smaller range. The degree of impact exerted by the aspect ratio is notably greater. Notably, mixing elements with different spin enhance shear and stretching efficiency. Compared to the same spin, SMD becomes 3.7–5.8 times smaller in the smaller size range with a significantly narrower distribution. Taking into account the pressure drop and efficiency in a comprehensive manner, optimized structural parameters for the mixing element encompass an aspect ratio of 1–1.5, a blade twist angle of 180°, an assembly angle of 90°, and interlaced assembly of adjacent elements with different spin. This work provides vital theoretical underpinning and future reference for enhancing KSM performance.

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1. Introduction

Liquid–liquid dispersion systems find extensive application within the chemical industry for chemical unit operations such as extraction, polymerization and emulsification, which play a significant role across sectors such as food, cosmetics, minerals and petrochemicals. Notable dispersion equipment includes stirred tanks, high-pressure homogenizers, and static mixers. In contrast to traditional stirred tanks, static mixers are widely used in fine chemical, pharmaceutical, polymerization, and biochemical sectors

due to their brilliant mixing performance, rapid and consistent mixing, adjustable residence time, and energy-efficient power homogenization [1–5]. Owing to the intricate interplay of various external and internal factors, including mixing elements, radial secondary flows, fluid turbulence, droplet aggregation and breakage, interphase mass transfer and dynamic interphase coupling, the liquid–liquid flow and mixing behavior within static mixers manifest remarkable complexity. Consequently, studying the effects of different operating conditions and internal structural parameters of static mixers is crucial for achieving efficient mixing.

The initial exploration of dispersion performance in static mixers is conducted through experimental methods. Middleman [6] investigated six low-viscosity ($\mu_d < 0.03$ Pa·s) organic fluids as dispersed phases in Kenics static mixers (KSM) under turbulent conditions. He formulated an expression grounded in Weber and

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Reynolds numbers in relation to the Sauter mean diameter (SMD): $d_{32}/D = c_1 We^{c_2} Re^{c_3}$. Grace [7] delved into the liquid–liquid dispersive kinetics in static mixers and examined droplet deformation principles under various conditions and the creeping flow of a wide range of viscosity ratios ($10^{-6} < r_{vis} < 950$) and of high continuous phase viscosities ($0.5 < \mu_c < 300 \text{ Pa}\cdot\text{s}$). The research systematically investigated the effects of SMD on wall shear rate and hydraulic power under different inlet conditions, mixer geometries, viscosity ratios, interfacial tension and dispersed phase volume fractions. Berkman and Calabrese [8] studied the impact of dispersed phase viscosity on turbulent dispersion of viscous oil in KSM. Drawing from experimental findings, a semi-empirical theory that related average droplet size to Weber number and normalized SMD of d_{32} was established, yielding a correlation between droplet size distribution (DSD) and cumulative volume frequency. Because of the uneven distribution of each phase in the static mixer, the experiment cannot capture finer details accurately. The traditional experimental method to gauge the flow field and DSD in the static mixer proved complicated and costly. Moreover, deciphering the intricate relationship between them mechanistically presented its challenges.

With the progression of computer technology and computational fluid dynamics (CFD), numerical simulation has become an important alternative to experimental research [9–13]. Early numerical simulations focused on the fluid dynamic behavior of single-phase flow in different static mixers. Hobbs *et al.* [14] used CFD to simulate the flow of low Reynolds number in KSM, investigating the thickness of the mixing element in static mixers and the transition between two elements. The fluid flow characteristics in KSM were analyzed, and the simulated pressure drop data was cross-referenced with multiple experimental correlations extracted from prior literature about pressure drops within KSM. Song and Han [15] formulated a pressure drop correlation for KSM using CFD, encompassing the entire Reynolds number spectrum from laminar to turbulent flow and standing distinct from earlier ones. van Wageningen *et al.* [16] employed both CFD and lattice Boltzmann methods to simulate the flow distribution of the transitional flow in KSM, comparing the results with laser Doppler anemometry measurements. Although previous CFD studies have yielded achievements, there persists a need to employ more fitting models for subsequent simulation computations, aimed at minimizing model inaccuracies and achieving a higher level of intricate insight.

To investigate the dynamic distribution of droplet size and the phenomena of droplet breakage and aggregation, many scholars have employed the computational fluid dynamics-population balance model (CFD-PBM) coupled method for systematic studies [17–19]. Farzi *et al.* [20] conducted CFD-PBM simulations of liquid–liquid dispersion in KSM with Reynolds numbers below 800. The simulation results revealed that the initial droplet position strongly influenced the droplet breakage probability, thus affecting the DSD. Meng *et al.* [21] numerically simulated turbulent dispersion and mixing of water–silicone oil two-phase flow in a Q-type static mixer using the CFD-PBM coupled model. The findings indicated that the dimensionless d_{32} decreased with an increase of the mixing element number and Reynolds number and stabilized below 0.03 at axial length of $z/l > 10$. The study also extracted empirical correlations between SMD, Weber number, and residence time to predict the equilibrium characteristics of droplet groups. Jaworski *et al.* [22] investigated turbulent droplet breakage in KSM by solving the transport equation for DSD moments. Haddadi *et al.* [17] used the Euler–Euler and Euler–Lagrangian methods to simulate turbulent liquid–liquid dispersion in KSM. The simulation results showed that as the Weber number and the number of mixing

elements increased, the SMD decreased, and the residence time decreased with an increase in the Weber number or a decrease in the number of mixing elements. Azizi and Al Taweel [23] employed CFD-PBM simulations to explore turbulent liquid–liquid dispersion in a screen-type static mixer. The numerical results demonstrated that the accuracy of the Coulaloglou and Tavlarides (CT) breakage and aggregation kernels in predicting droplet behavior within the screen-type static mixer. Currently, most studies focus on the influence of the number of mixing elements on DSD and liquid–liquid dispersion processes in different static mixers. However, a notable gap remains in terms of systematic analysis and investigations on specific structural parameters of mixing elements, which are crucial for the structural optimization and process enhancement of KSM. Furthermore, it is proposed to use the discrete method to solve the population balance equation (PBE), which can provide more accurate and finer-scale simulation results compared to conventional moment–based methods.

Therefore, based on the CFD-PBM coupled method and after validating the model accuracy, this work systematically investigates the effects of operating conditions and structural parameters of mixing elements on liquid–liquid dispersion characteristics within KSM, including the aspect ratio, blade twist angle, blade assembly angle, and swirl direction of blade. The focus lies on meticulous analysis of DSD and SMD in liquid–liquid dispersion systems, revealing dispersion characteristics and mechanisms. This work provides essential theoretical support and reference for the structural optimization and process enhancement of KSM in the future.

2. Mathematic Formulation

2.1. Eulerian model

Euler–Euler model is effective for simulating dispersion in liquid–liquid multiphase systems. According to the Euler–Euler model, both phases are treated as continuum interpenetrating phase, incorporating the concept of phase volume fraction. Meanwhile, the momentum equation and the mass conservation equation were solved for each phase.

The continuity equation is given as follows:

$$\frac{\partial}{\partial t}(\alpha_q \rho_q) + \nabla \cdot (\alpha_q \rho_q \mathbf{u}_q) = 0 \quad (1)$$

Momentum balance equation

$$\frac{\partial}{\partial t}(\alpha_q \rho_q \mathbf{u}_q) + \nabla \cdot (\alpha_q \rho_q \mathbf{u}_q \mathbf{u}_q) = -\alpha_q \nabla p + \nabla \cdot (\alpha_q \boldsymbol{\tau}_q) + \alpha_q \rho_q \mathbf{g} + \mathbf{S}_i \quad (2)$$

Volume fraction equation

$$\sum_{q=1}^n \alpha_q = 1 \quad (3)$$

where, q denotes the phase sequence number. α , ρ , $\mathbf{u}$, p and $\mathbf{g}$ denote the volume fraction, density, velocity, pressure share by two phase and gravity acceleration vector. $\boldsymbol{\tau}$ denotes the stress-strain tensor, which includes the viscous stress tensor and the Reynolds number stress tensor. $\mathbf{S}_i$ denotes the source term of interfacial momentum exchange, which includes the effects of drag, lift and virtual mass forces. In this work, only the drag force $\mathbf{F}$ is considered.

$$\mathbf{F} = \alpha_{dis} \alpha_c \left(\frac{3}{4} C_{drag} \frac{\rho_c}{d} |\mathbf{u}_{dis} - \mathbf{u}_c| \right) (\mathbf{u}_{dis} - \mathbf{u}_c) \quad (4)$$

$$\mathbf{F}_{\text{drag,dis}} = -\mathbf{F}_{\text{drag,c}} \quad (5)$$

where, c and dis represent continuous phase and dispersed phase, respectively. In this work, C_{drag} represents the drag coefficient, which is estimated by the Schiller and Naumann. It applies to the liquid–liquid two–phase systems.

$$C_{\text{drag}} = \begin{cases} \frac{24}{Re} (1 + 0.15Re^{0.687}), & Re_r \leq 1000 \\ 0.44, & Re_r > 1000 \end{cases} \quad (6)$$

$$Re_r = \frac{\rho_c |\mathbf{u}_c - \mathbf{u}_d| d}{\mu_c} \quad (7)$$

where, Re_r denotes the relative Reynolds number and d denotes the local droplet diameter. Since droplet diameters are spatially distributed in KSM, predicting interfacial momentum based on a single droplet size is imprecise. To overcome this shortcoming, the spatial distribution of droplet size is determined by solving the PBM.

2.2. Turbulence model

2.2.1. Standard k – ε model

Standard k – ε model [24] is a model of the transport equation based on turbulent kinetic energy k and energy dissipation rate ε . The transport equation for the turbulent kinetic energy k is derived from the exact equation, while the transport equation for the energy dissipation rate ε is obtained by physical reasoning and bears little resemblance to the mathematically exact equation. The transport equations are given as follows:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \varepsilon - Y_M + S_k \quad (8)$$

$$\frac{\partial}{\partial t}(\rho \varepsilon) + \frac{\partial}{\partial x_i}(\rho \varepsilon u_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + C_{1\varepsilon} \frac{\varepsilon}{k} (G_k + C_{3\varepsilon} G_b) - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k} + S_\varepsilon \quad (9)$$

where, $\mu_t = \rho C_\mu k^2 / \varepsilon$, $C_{1\varepsilon} = 1.44$, $C_{2\varepsilon} = 1.92$, $C_\mu = 0.09$, $\sigma_k = 1.0$, $\sigma_\varepsilon = 1.3$.

2.2.2. Standard k – ω model

Standard k – ω model is an empirical model of the transport equation based on the turbulent kinetic energy k and the specific dissipation rate ω , which can also be treated as a ratio of k to ω . The transport equations are given as follows:

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_j} \left(\Gamma_k \frac{\partial k}{\partial x_j} \right) + G_k - Y_k + S_k + G_b \quad (10)$$

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_i}(\rho \omega u_i) = \frac{\partial}{\partial x_j} \left(\Gamma_\omega \frac{\partial \omega}{\partial x_j} \right) + G_\omega - Y_\omega + S_\omega + G_{\omega b} \quad (11)$$

where, G_k denotes the turbulent kinetic energy due to the mean velocity gradient. G_ω denotes the generation of ω . $G_{\omega b}$ represents the

generation of ω . Γ_k and Γ_ω represent the effective diffusivity of k and ω , respectively. Y_k and Y_ω represent the k and ω dissipation from turbulence. S_k and S_ω are user–defined source code terms. G_b and $G_{\omega b}$ explain the buoyancy terms described in the Standard k – ω model of the effect of buoyancy on turbulence.

The effective diffusion coefficient is calculated as follows:

$$\Gamma_k = \mu + \frac{\mu_t}{\sigma_k} \quad (12)$$

$$\Gamma_\omega = \mu + \frac{\mu_t}{\sigma_\omega} \quad (13)$$

where, σ_k and σ_ω denote the turbulent Prandtl numbers for k and ω , respectively.

The turbulent viscosity is calculated as follows:

$$\mu_t = \alpha^* \frac{\rho k}{\omega} \quad (14)$$

$$\alpha^* = \alpha_\infty^* \left(\frac{\alpha_0^* + Re_t / R_k}{1 + Re_t / R_k} \right) \quad (15)$$

$$Re_t = \frac{\rho k}{\mu \omega} \quad (16)$$

where, $R_k = 6$, $\alpha_0^* = \beta_i / 3$, $\beta_i = 0.072$.

2.2.3. SST k – ω model

In three-dimensional flows with complex geometries [25], the SST k – ω model shows better performance than the Standard k – ω model. The SST k – ω model has a good predictive ability in complex turbulent systems, considering the empirical numerical case for the selection of the two-equation turbulence model. This turbulence model utilizes the transport equation with specific turbulent dissipation rate ω and turbulent kinetic energy k , the equations are given as follows:

$$\nabla \cdot (\alpha_c \rho_c k_c \mathbf{u}_c) = \nabla \cdot \left[\left(\mu_c + \frac{\mu_t}{\sigma_{k,c}} \right) \nabla k_c \right] + \alpha_c (G_{k,c} + G_{kb,c} + S_{k,c} - Y_{k,c}) \quad (17)$$

$$\nabla \cdot (\alpha_c \rho_c \omega_c \mathbf{u}_c) = \nabla \cdot \left[\left(\mu_c + \frac{\mu_t}{\sigma_{\omega,c}} \right) \nabla \omega_c \right] + \alpha_c (G_{\omega,c} + G_{\omega b,c} + S_{\omega,c} - Y_{\omega,c}) \quad (18)$$

where, $\sigma_{k,c}$ and $\sigma_{\omega,c}$ denote the turbulence Prandtl number for k and ω , respectively; Y_k and Y_ω are the dissipation rates due to turbulence for k and ω ; μ_t denotes the turbulent eddy viscosity.

The turbulent eddy viscosity μ_t is given by:

$$\mu_t = \frac{\rho k_c}{\omega_c} \frac{1}{\max(1/\alpha^*, SF_1/a_1 \omega_c)} \quad (19)$$

$$F_1 = \tanh(\varphi_1^4) \quad (20)$$

$$\varphi_1 = \max \left(\frac{2\sqrt{k_c}}{0.09\omega_c y}, \frac{500\mu_c}{\rho y^2 \omega_c} \right) \alpha^* = \alpha_\infty^* \left(\frac{\beta_i/3 + Re_t/R_k}{1 + Re_t/R_k} \right) \quad (21)$$

where, F_1 and φ_1 represent closure coefficients, y represents the distance to the nearest surface, $Re_t = \rho k / \mu_t \omega$, and β_i , R_k and α_∞^* represent model constants.

2.3. Population balance model

2.3.1. Population balance equation

Combining the dispersed-phase droplet distribution with microscopic behavior based on population balance theory, the number density transport equation of the droplet can be established for liquid–liquid two-phase flow systems. In the ANSYS Fluent, the PBE is solved by finite volume method, and the transport equation of number density function (NDF) as follows:

$$\frac{\partial n(d, t)}{\partial t} + \nabla \cdot [n(d, t) \mathbf{U}] = B_{br} - D_{br} + B_{ag} - D_{ag} \quad (22)$$

where, $n(d, t)$ denotes NDF, meaning the number concentration of droplets of diameter d at time t . $\mathbf{U}$ denotes the velocity of droplet, B_{br} and B_{ag} represent the rate of droplet birth due to breakage and aggregation, respectively, D_{br} and D_{ag} represent the rate of droplet death due to breakage and aggregation. The expressions for the rate of droplet birth and death are given as follows:

$$B_{br} = \int_0^{+\infty} c \Gamma(d') n(d', t) \beta(d, d') dd' \quad (23)$$

$$D_{br} = \Gamma(d) n(d, t) \quad (24)$$

$$B_{ag} = \frac{d^2}{2} \int_0^d \frac{h\left((d^3 - d'^3)^{\frac{1}{3}}, d'\right)}{(d^3 - d'^3)^{\frac{2}{3}}} n\left((d^3 - d'^3)^{\frac{1}{3}}\right) n(d') dd' \quad (25)$$

$$D_{ag} = \int_0^{+\infty} h(d, d') n(d, t) n(d', t) dd' \quad (26)$$

where, $\Gamma(d)$ and $\beta(d, d')$ represent the droplet breakage frequency and daughter droplet diameter distribution function, respectively. c is the number of daughter droplet formed by a breakup droplet. $h(d, d')$ is the aggregation kernel, representing the product of the collision frequency and the aggregation efficiency of dispersed droplets.

2.3.2. Turbulent aggregation kernel

In our work, the turbulent model is applied to describe the aggregation behavior of droplets based on film drainage theory. In the turbulent model, aggregation occurs through two mechanisms: (1) The viscous subrange mechanism is used when the droplets are smaller than the Kolmogorov scale η and (2) The inertial subrange mechanism is used when the droplet is larger than the Kolmogorov scale η [21,26]. In the ANSYS Fluent, turbulent model consists of the above two mechanisms.

For the viscous subrange, the collision rate is expressed in Eq. (27).

$$a(d_i, d_j) = \zeta_T \sqrt{\frac{8\pi}{15}} \gamma \frac{(d_i + d_j)^3}{8} \quad (27)$$

For the inertial subrange, droplet is larger than the smallest vortex, the collision rate is expressed by Abrahamson's model [27].

$$a(d_i, d_j) = \zeta_T 2^{3/2} \sqrt{\pi} \frac{(d_i + d_j)^2}{4} \sqrt{(U_i^2 + U_j^2)} \quad (28)$$

where, ζ_T denotes the pre-factor that takes the capture efficiency coefficient of turbulent collision in account, γ denotes the shear rate and U_i^2 denotes the mean-square velocity of droplet.

2.3.3. Breakage kernel

Different breakage kernels have been put forward to describe the breakage of droplets. Table 1 illustrates several popular models.

2.3.4. The discrete method

Nowadays, the PBE is solved by the method of moments in most of the relevant studies. The method involves solving the NDF by transforming it into the transport equations, but its accuracy relies on the method of closed transport equations, and it is not straightforward to obtain the NDF. However, the discrete method is used to solve the population balance equation, which is able to directly discretize the NDF into a large number of droplet levels as a way to represent the continuous droplet size distribution [31]. Therefore, more accurate simulation results can be obtained. The discrete method has significant advantages such as good robustness and easy access to DSD. The discrete method is that the droplets are classified according to the same volume ratio as each other. It is assumed that the droplets are spherical, and the volume ratio of each class is shown in Eq. (29).

$$\frac{V_i}{V_{i+1}} = 2^q \quad (29)$$

where, q represents any integer greater than or equal to 1.

3. Numerical Simulations

3.1. Geometric model

In this work, the KSM is based on the experiment of Berkman and Calabrese [8], and the geometry is shown in Fig. 1. The KSM has a concentric inlet consisting of a continuous phase inlet and a dispersed phase inlet with a diameter of $D_0 = 2.5$ mm. The diameter of the tube is 19.1 mm, and the length of the tube is 493.8 mm including inlet and outlet sections that are 75 mm. The thickness of the mixing element is 3.3 mm, and the length of the mixing element assembled with a rotation angle of 90° is 28.65 mm.

3.2. Physical properties

This work used water as the continuous phase and silicone oil as the dispersed phase, and the volume fraction of the dispersed phase is 0.001–0.05. Physical properties of fluids are shown in Table 2.

3.3. Boundary conditions

The commercial software ANSYS Fluent was used for the numerical simulations, which was based on the pressure-base solver. Water was set as primary phase, while silicone oil was set as secondary phase. The liquid–liquid interphase force was only considered as the drag force, which was calculated by the Schiller and Naumann model. The discrete method was adopted to solve the PBE. The liquid–liquid two phase inlets were defined as velocity boundary condition, and the outlet boundary condition was set as outflow. No slip boundary condition was employed to the mixing

Table 1
Comparison of breakage kernels

Researchers	Breakage kernel function
Coulaloglou and Tavlarides [28] Breakage kernel	$\Omega(d) = c_1 \frac{\varepsilon^{1/3}}{(1 + \alpha_d)d^{2/3}} \exp\left(-c_2 \frac{\sigma(1 + \alpha_d)^2}{\rho_d \varepsilon^{2/3} d^{5/3}}\right)$
Daughter droplet diameter distribution function	$\beta(d, d') = 14.4 \frac{d^2}{d'^3} \exp\left(-4.5 \frac{2d^3 - d'^3}{d'^6}\right)$
Lehr et al. [29] Breakage kernel	$\Omega_{br}(d, d') = K \int_{\xi_{min}}^1 \frac{(1 + \xi)^2}{\xi^h} \exp(-b\xi^m) d\xi$ $K = 1.19e^{-1/3} d^{-7/3} \sigma \rho^{-1} f^{-1/3}, b = 2We_{crit} \sigma \rho^{-1} e^{-2/3} d^{-5/3} f^{-1/3}$
Daughter droplet diameter distribution function	$\beta(d, d') = \frac{1}{\sqrt{\pi} f_{BV}} \cdot \frac{\exp\left\{-\frac{9}{4} \left[\ln\left(\frac{2^{2/5} d' \rho_c^{3/5} \varepsilon^{2/5}}{\sigma^{3/5}}\right)\right]^2\right\}}{1 + \operatorname{erf}\left[\frac{3}{2} \ln\left(\frac{2^{1/5} d' \rho_c^{3/5} \varepsilon^{2/5}}{\sigma^{3/5}}\right)\right]}$ $\Omega_{br}(d, d') = K \int_{\xi_{min}}^1 \frac{(1 + \xi)^2}{\xi^h} \exp(-b\xi^m) d\xi$
Luo and Svendsen [30] Breakage kernel	$\Omega_{br}(d, d') = K \int_{\xi_{min}}^1 \frac{(1 + \xi)^2}{\xi^h} \exp(-b\xi^m) d\xi$ $K = 0.9238(1 - \alpha_d)e^{1/3} d^{-2/3}, c_f = f_{BV}^{2/3} + (1 - f_{BV})^{2/3} - 1, b = -\frac{12c_f \sigma}{\beta \rho_c \varepsilon^{2/3} d^{5/3}}$
Daughter droplet diameter distribution function	$\beta(d, d') = \frac{2 \int_{\xi_{min}}^1 (1 + \xi)^2 \xi^{-11/3} e^{-\chi_c} d\xi}{\int_0^1 \int_{\xi_{min}}^1 (1 + \xi)^2 \xi^{-11/3} e^{-\chi_c} d\xi df_{BV}}$ $\chi_c = \frac{12c_f \sigma}{\beta \rho_c \varepsilon^{2/3} d^{5/3} \xi^{11/3}}$

element wall and the tube wall. The pressure-velocity coupling was achieved by Phase Coupled SIMPLE algorithm. The least squares cell-based method was adopted to solve grids gradient. The PRES-TO! scheme was used to calculate pressure discretization. The first order upwind scheme was selected for the momentum equation, while the QUICK scheme was adopted to calculate other equations. The time step employed in the simulations was set 0.002 s. The residual for all equations was 10^{-6} except the continuity equation which had a residual of 10^{-5} . The simulations were performed on a workstation with an Intel Xeon Gold 5218R CPU@2.10 GHz and 128 GB of RAM.

4. Results and Discussion

4.1. Model verification

4.1.1. Selection of turbulence models and breakage models

The selection of turbulence models and breakage models greatly affects the accuracy of CFD calculations. In this work, the Standard $k-\varepsilon$ model, Standard $k-\omega$ model, and SST $k-\omega$ model are used as turbulence models, while the CT model [28], Lehr model [29], and Luo model [30] are used as breakage models for numerical simulations. The simulation results are compared with experimental results.

As shown in Fig. 2(a), the SMD obtained using the SST $k-\omega$ model closely matches the experimental values from Berkman and Calabrese [8], with an average relative error of 3.06%. Contrarily, the Standard $k-\varepsilon$ model yields an underprediction of the Sauter mean diameter, while the Standard $k-\omega$ model produces an overestimation. As shown in Fig. 2(b), the simulated values of the Luo model and CT model show a gradual decrease with escalating Reynolds number, while the simulated values of Lehr model ($We_{crit} = 0.1$) display a slight increase with Reynolds number increments and the simulated values of the Lehr model ($We_{crit} = 1$)

initially exhibit a declining tendency, followed by a slight increase as Reynolds number continues to rise. The simulated values of Luo model, CT model, Lehr model ($We_{crit} = 1$), and Lehr model ($We_{crit} = 0.1$) demonstrates average relative errors of 4.62%, 23.5%, 20.6% and 65.3%, respectively. The divergence is due to the fact that Luo model, Lehr model, and CT model share the isotropic turbulence theory [29], but hold different interpretations of turbulence-induced droplet breakage. Luo model postulates droplet breakage as a binary process, suggesting that the turbulent kinetic energy of turbulence vortex exceeding the increment of surface energy before and after the breakage leads to droplet breakage. Lehr model considers the impact of turbulence and buoyancy on collisions, suggesting that the inertial force generated by turbulence vortex must surpass the interfacial tension of the smallest droplet following the breakage, thereby inducing the breakage [32]. In turbulent liquid–liquid dispersion systems, the critical Weber number for droplet breakage is about 1 [17]. CT model assumes both phases move at identical velocities, proposing that the turbulent kinetic energy of turbulence vortex must exceed the droplet's surface energy to induce breakage. Hence, SST $k-\omega$ model is selected to the turbulence model and Luo model is used to the breakage model.

4.1.2. Comparison between numerical simulation and experiment

To validate the predictive capability of the CFD-PBM coupled model, the numerical results of water–oil two-phase mixing characteristics in KSM were compared with experimental results. The variation of SMD with different Reynolds numbers is shown in Fig. 3. As the Reynolds number increases, turbulence intensity enhances, leading to a decrease in the SMD. For the SMD, the relative error range between numerical simulation results and Berkman and Calabrese's [8] experimental results is 2.6%–10.8%, with an average relative error of 4.6%, which proved superior to the 6.5% from Jaworski et al. [22]. Moreover, as the Reynolds number

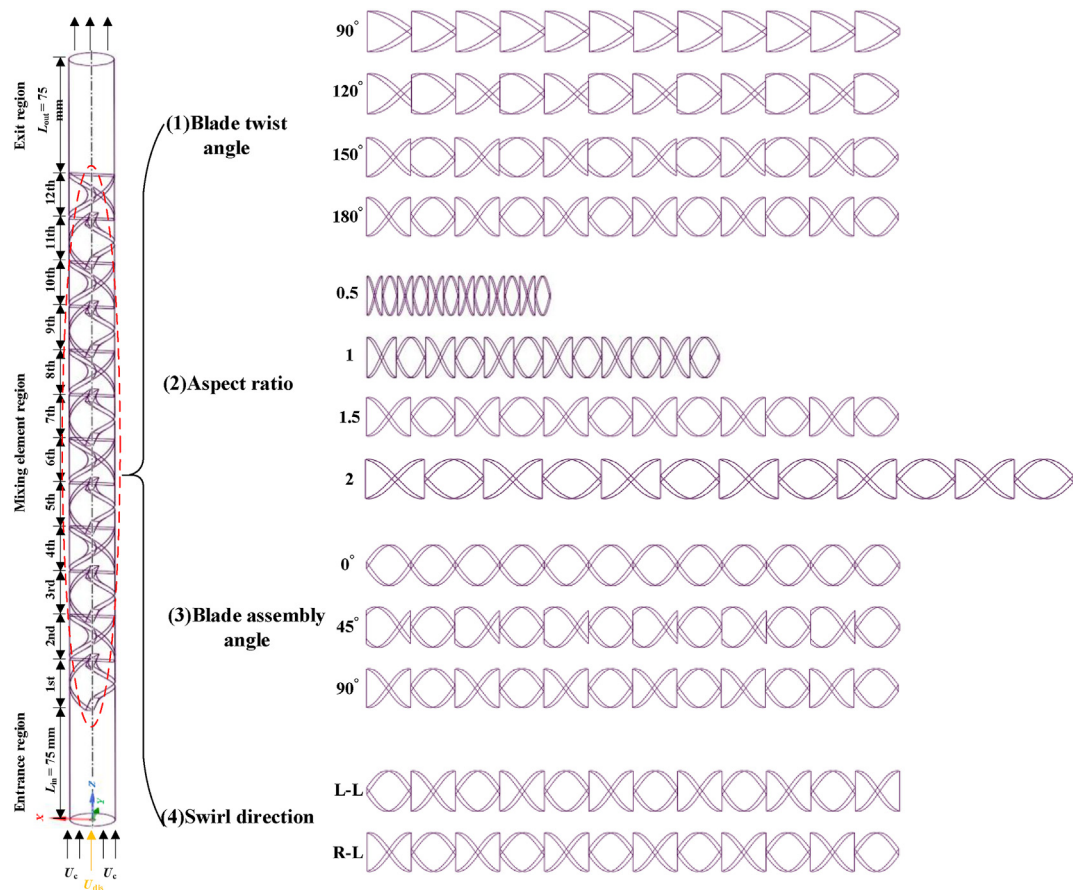


Fig. 1. The geometric structure of Kenics static mixer.

Table 2
Physical properties of fluids

Fluid	Density/kg·m ⁻³	Viscosity/Pa·s	Interfacial tension/N·m ⁻¹
Water	998.2	0.001003	—
Silicone oil	963	0.0518	0.0374

increases, the simulated results exhibit a convergence towards the experimental results, indicating an improved accuracy of the simulation.

4.1.3. Grid independence verification

Grid independence verification is shown in Fig. 4. From Fig. 4(a), it can be observed that as the grid number increased, the SMD decreases and stabilizes, while the energy dissipation rate increases and stabilizes. As the grid number exceeds 946169, both the SMD and the energy dissipation rate show no significant changes. Fig. 4(b) illustrates the pattern of continuous phase velocity distribution at the outlet of the KSM, which conforms to the distinctive “M” shape [21]. As the grid number escalates, the “M” shape distribution curve of continuous phase velocity becomes progressively

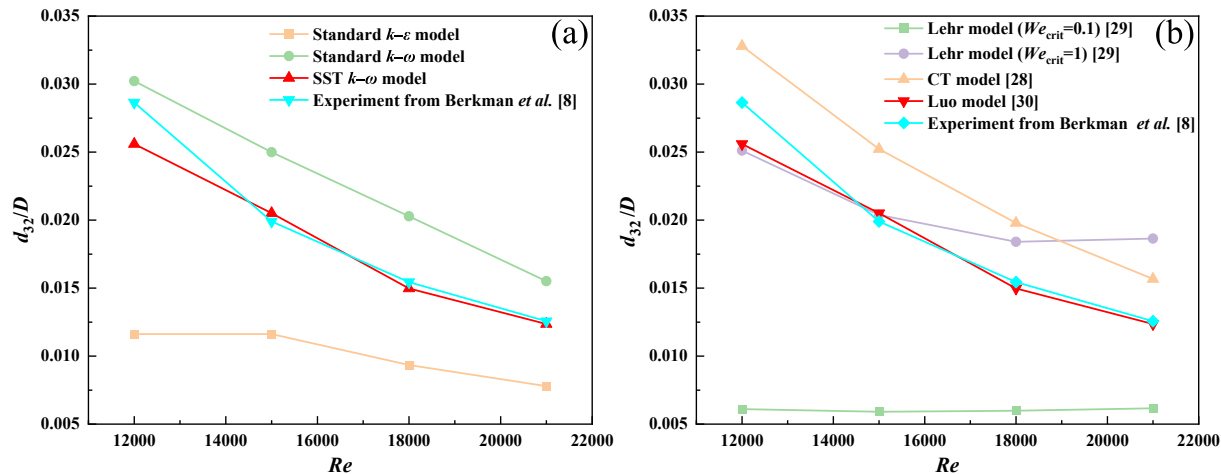


Fig. 2. Effects of turbulence models and breakage models on the SMD at different Reynolds numbers ($\alpha_d = 0.001$): (a) turbulence models, (b) breakage models.

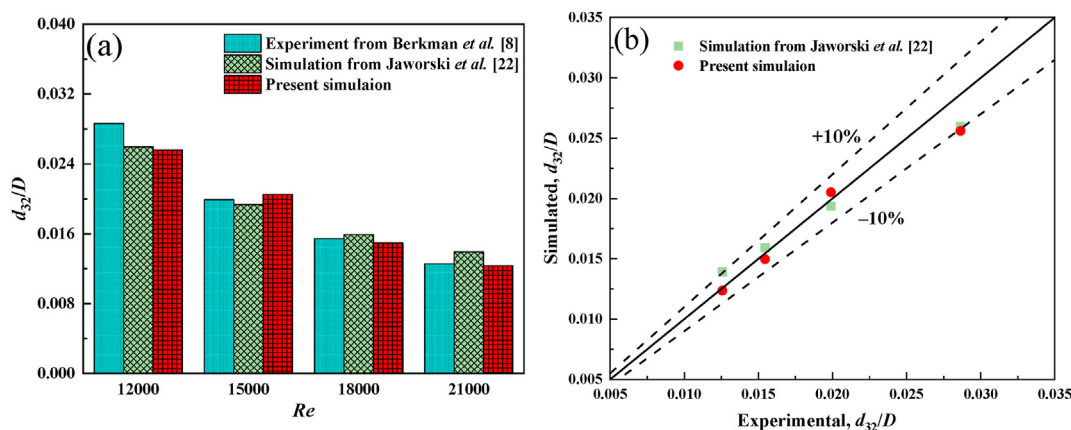


Fig. 3. Model validation (breakage model: Luo model): (a) Sauter mean diameter of different Reynolds numbers, (b) quantitative comparison between simulated and experimental values.

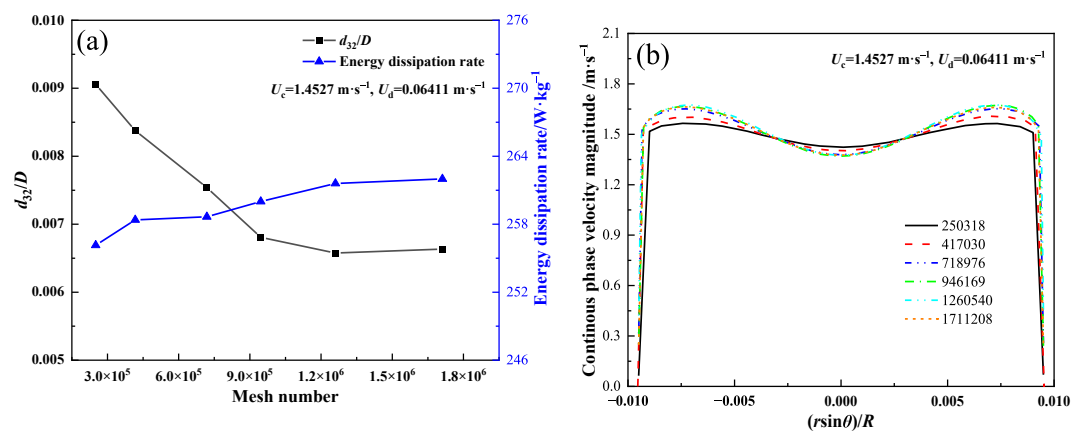


Fig. 4. Grid independence tests: (a) Sauter mean diameter and energy dissipating rate, (b) continuous phase velocity.

smoother. When the grid number surpassed 946169, the distribution of continuous phase velocity maintains a relatively steady state. Taking into consideration both computational efficiency and accuracy, the grid number of 946169 is selected for the model calculations.

4.1.4. Determination of bin number

To determine an appropriate number of bin and the droplet size range, the dispersed phase droplet sizes are divided into 8, 12 and 16 bins within the droplet size range of 0.002–5 mm in this study. From Fig. 5(a), it can be observed that in the inlet section of the KSM, there are notable differences in the SMD between 8, 12 and 16 bins. However, the differences become less significant after the fluid enters the mixing element section. This is due to the substantial difference in droplet size ranges between 8 bins (0.015–5.067 mm), 16 bins (0.004–2.506 mm), and 12 bins (0.002–2.5 mm), leading to distinct differences about SMD in the inlet section. This conclusion is also evident in the DSD. DSD is the droplet size distribution, which is displayed as the number density of each droplet group around size d_i . Fig. 5(b) indicates that irrespective of the bin number, the droplets of the dispersed phase spanning the range of 0.015–0.19 mm contribute most significantly to the overall droplet population. In summary, droplets larger than 2.50 mm have minimal impact on the simulation results. Considering both simulation efficiency and accuracy,

this work has opted for 12 bins for the dispersed phase droplet size within the range of 0.004–2.506 mm.

4.2. Operating conditions

4.2.1. Reynolds number

The influence of Reynolds number on droplet size and DSD is depicted in Fig. 6. As shown in Fig. 6(a), with an increase in Reynolds number, turbulence intensity enhances and the energy dissipation rate increases, which provides more energy to vortices, making droplets more prone to breakage and leading to a reduction in the SMD. In the inlet section of the KSM, dispersed phase droplets hardly undergo breakage with the droplet size of about 2.48 mm. This phenomenon stems from the inadequacy of turbulence energy dissipation rate to furnish the requisite energy for droplet breakage. Hence, the dispersed phase within this section scarcely experiences significant breakage or size reduction. As the axial flow distance increases, a succession of “segmentation–swirl–aggregation” unfolds under the influence of shear effects and turbulence vortices. This sequence makes droplets prone to deformation and breakage, ultimately resulting in smaller droplets. After the dispersed phase flows out of the mixing elements, droplets slightly enlarge. Fig. 6(b) shows that as the Reynolds number increases, the average droplet size decreases, the DSD becomes narrower, and it gradually shifts towards smaller size. Fig. 6(c) illustrates that droplets consume energy during breakup.

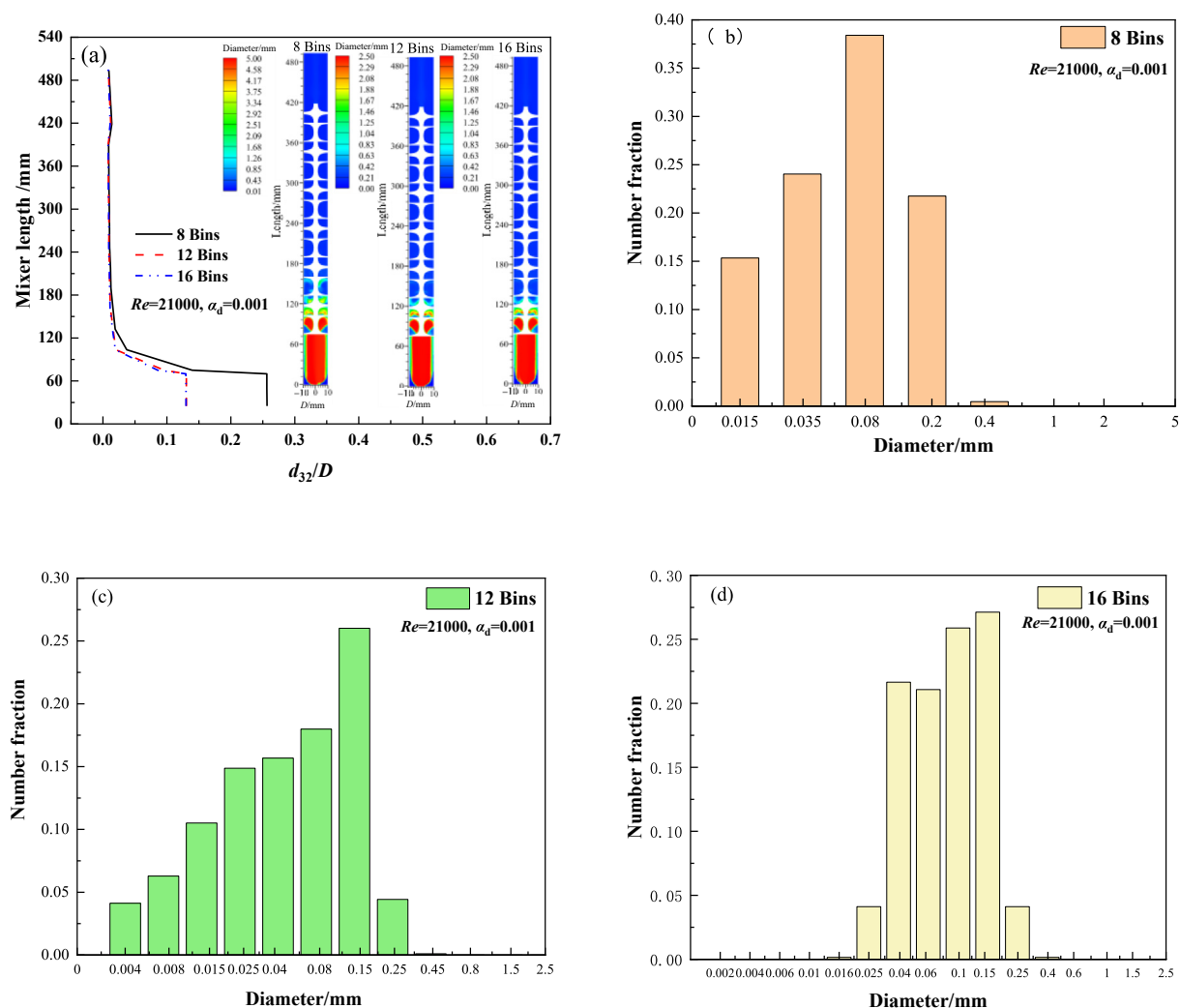


Fig. 5. Effects of bin number on simulation results: (a) Sauter mean diameter axial distribution, (b) droplet size distribution of 8 bins, (c) droplet size distribution of 12 bins, (d) droplet size distribution of 16 bins.

As the Reynolds number increases, turbulence is enhanced and the energy dissipation rate increases as the droplet size decreases. The effect of Reynolds number on dispersed phase volume fraction can be seen in Fig. S1 of Supplementary Material. The volume fraction figures of the other parameters be seen in Figs. S2–S6 and will not be repeated later.

4.2.2. Dispersed phase volume fraction

The effect of dispersed phase volume fraction on droplet size and DSD is illustrated in Fig. 7. From Fig. 7(a), it can be observed that in the inlet section of the KSM (0–75 mm), dispersed phase droplet sizes scarcely undergo breakage. As axial flow distance increases, fluid experiences divergence, confluence, rotation and turbulence-induced vortices, weakening axial mixing and promoting radial mixing for liquid–liquid dispersion. Therefore, as the flow distance of the dispersed phase increases, droplet sizes gradually reduce. After the dispersed phase flows out of the mixing elements, droplets are no longer subject to shear effect from the mixing elements. Turbulence intensity weakens, breakage rate decreases, and droplet size slightly increases. Furthermore, an augmentation in the dispersed phase volume fraction contributes to an increase in the SMD. This phenomenon finds its roots in two distinct factors. Primarily, as the volume

fraction of dispersed phase escalates, the energy required for droplet breakage increases due to the energy damping effect, which reduces the droplet breakage rate. Second, an elevation in the dispersed phase volume fraction augments the number of droplet groups, thereby enhancing droplet collision probability and aggregation rate [33]. Fig. 7(b) illustrates that as the dispersed phase volume fraction increases, breakage rate decreases, making droplets less prone to breakage and resulting in a broader droplet size distribution that gradually shifts towards larger size. Fig. 7(c) shows that as the volume fraction of the dispersed phase increases, the droplet size and the energy dissipation rate increase, which is detrimental to droplet breakup.

4.3. Structural parameters

4.3.1. Aspect ratio

The effect of aspect ratio on the SMD and droplet size is shown in Fig. 8. As can be seen from Fig. 8(a), the axial distribution pattern of SMD is similar for different aspect ratios. The larger the aspect ratio, the larger the SMD, and the dimensionless Sauter mean diameters at the outlet of the KSM for aspect ratios of 0.5, 1, 1.5 and 2 are 0.0060, 0.0106, 0.0140 and 0.0171, respectively. From

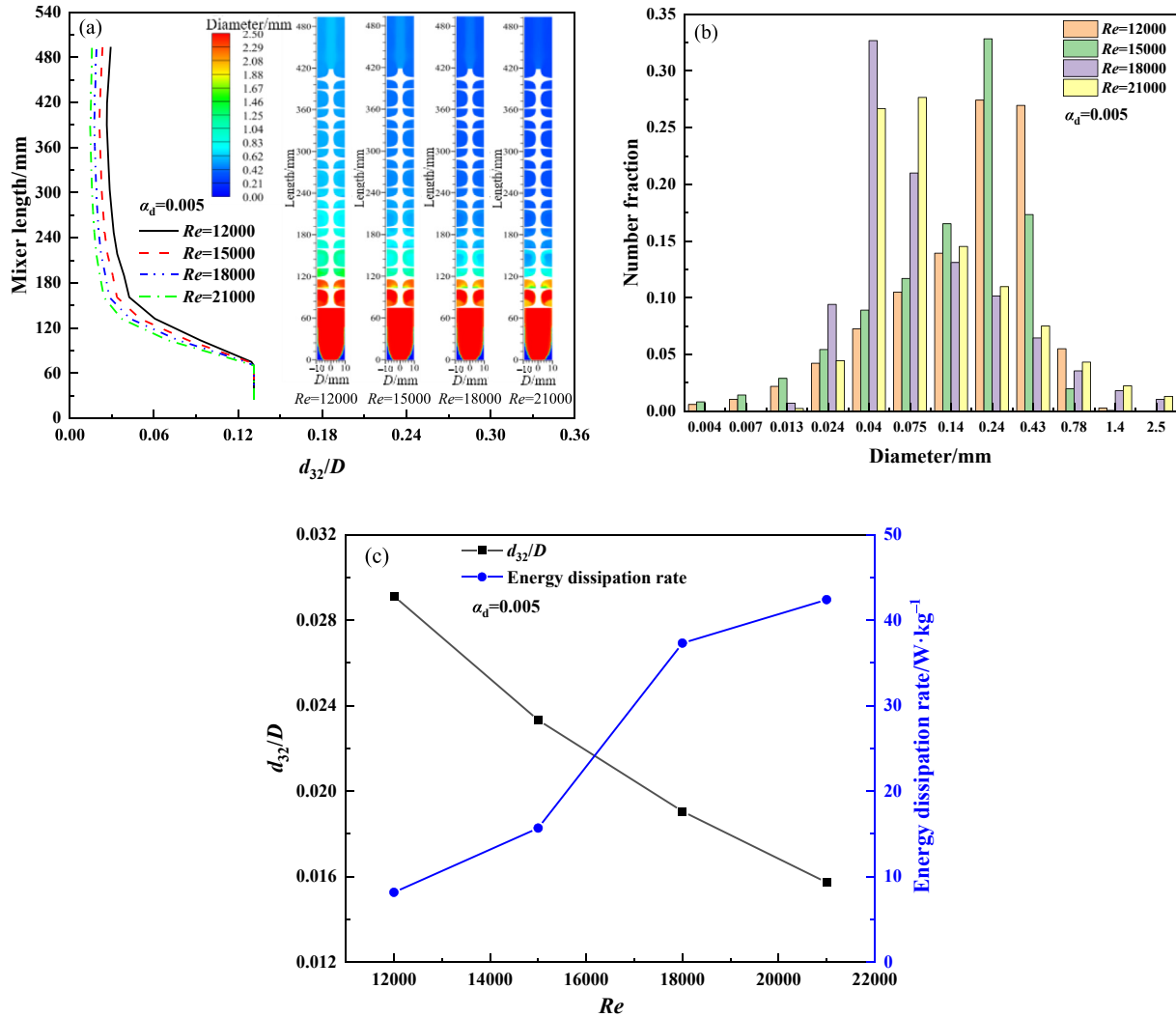


Fig. 6. Effects of Reynolds number on droplet size and its distribution ($\alpha_d = 0.005$): (a) Sauter mean diameter axial distribution, (b) droplet size distribution, (c) Sauter mean diameter and energy dissipation rate.

the distribution of the dispersion phase cloud diagram in Fig. 8(a), droplet sizes basically remain unchanged in the inlet section. After entering the mixing element section, the droplets are continuously subjected to the strong shear of the mixing element and the turbulent vortex induced by the mixing element, which leads to the droplet breakage until the average droplet size reaches the equilibrium state. The smaller aspect ratio results in rapid droplet breakage upstream of the mixing element, but this process is at the expense of more energy consumption. After the fluid flows through the first mixing element, the difference of SMD at different aspect ratios gradually appears. As the aspect ratio increases, the number of mixing elements required to stabilize the droplet size increases. For mixing elements with aspect ratios of 0.5, 1, 1.5 and 2, the droplet size basically ceases to change after the dispersion phase flow through the 3rd, 4th, 5th and 8th mixing elements. After the fluid flows out of the mixing elements, due to the weakening of turbulence, the mixing elements with larger aspect ratios generate vortices near this position as the velocity decreases and the droplet size increases. The turbulent dissipation rate and average energy dissipation rate for aspect ratios of 0.5, 1, 1.5 and 2 are 64.08 and 267.9 $m^2 \cdot s^{-3}$, 15.09 and 61.43 $m^2 \cdot s^{-3}$, 7.25 and 33.12 $m^2 \cdot s^{-3}$, 4.02 and 22.29 $m^2 \cdot s^{-3}$,

respectively. Combined with Fig. 8(b), at a small aspect ratio, the DSD is narrow, and the droplet size is small. As the aspect ratio increases, the energy dissipation rate decreases, which reduces the droplet breakage rate and is not conducive to droplet breakage. The DSD shifts towards the larger size.

The effect of aspect ratio on shear rate and stretching efficiency is shown in Fig. 9. The average shear rates of the mixed elements with aspect ratios of 0.5, 1, 1.5 and 2 are 3245.55, 1613.22, 1229.31 and 918.94 s^{-1} , and the average stretching efficiencies are 0.557, 0.553, 0.558 and 0.559, respectively. It can be seen in Fig. 9(a) that as the aspect ratio increases, the shearing action of the mixing elements on fluid at the same time is weakened, which also weakens the turbulent vortex and reduces the shear rate. At the position where the two mixing elements are at 90° interlaced junction, the shear rate reaches a peak.

The stretching efficiency is the ability of liquid to disperse droplets more uniformly and finely by stretching during dispersion and mixing process, which is used as the important parameter for dispersion and mixing process in the KSM. Fig. 9(b) shows that the stretching efficiency in the KSM shows a sawtooth wave trend. The stretching efficiency is larger in the upstream of the inlet section and the mixing element section of the KSM, and is relatively small

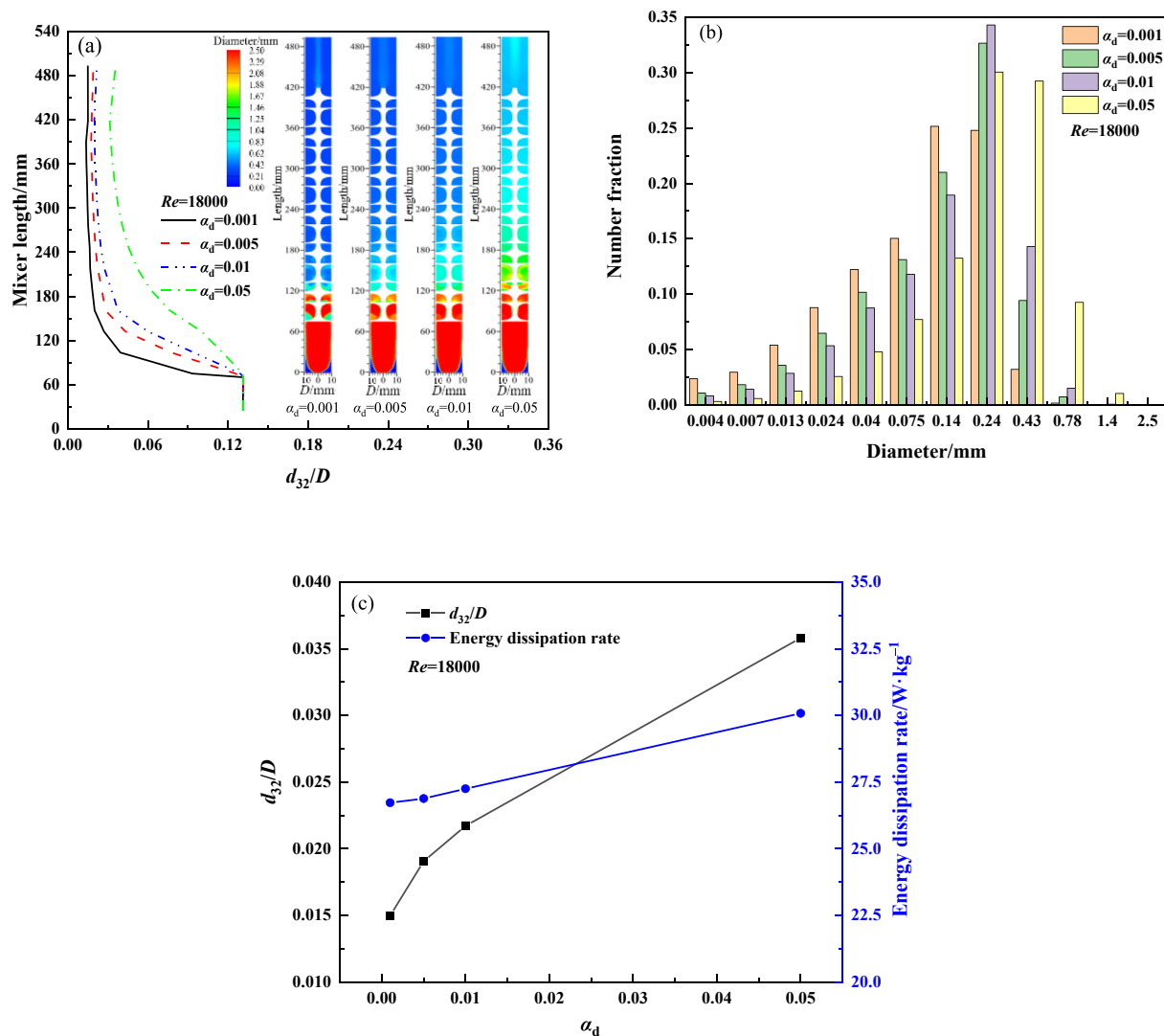


Fig. 7. Effects of dispersed phase volume fraction on droplet size and its distribution ($Re = 18000$): (a) Sauter mean diameter axial distribution, (b) droplet size distribution, (c) Sauter mean diameter and energy dissipation rate.

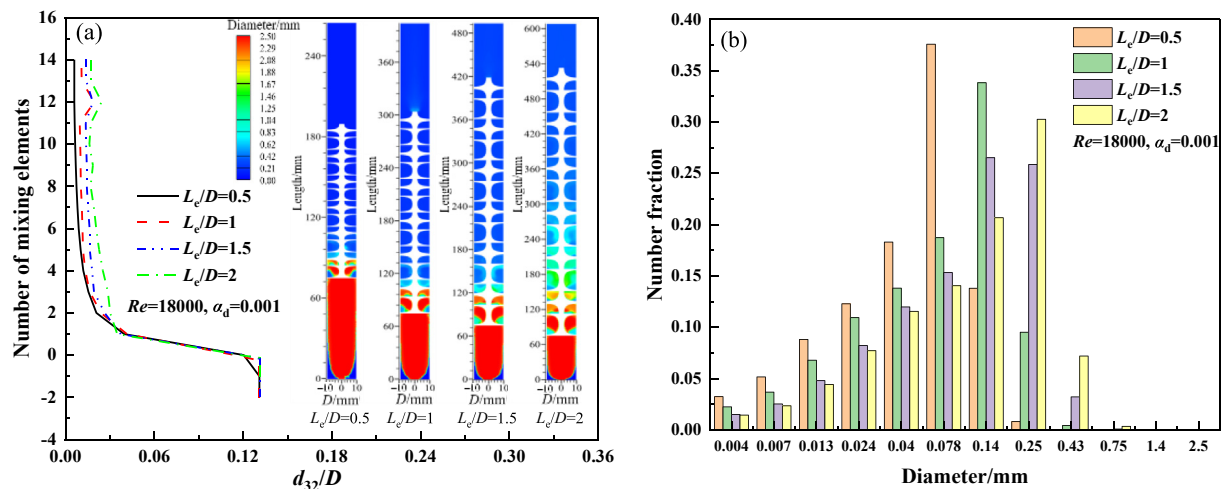


Fig. 8. Effects of aspect ratio on droplet size and its distribution: (a) Sauter mean diameter, (b) droplet size distribution.

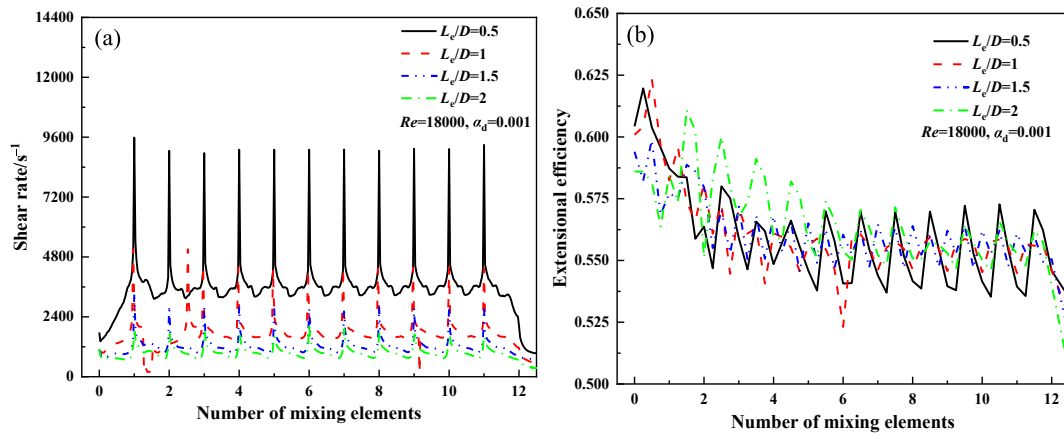


Fig. 9. Effects of aspect ratio on shear rate and stretching efficiency: (a) shear rate, (b) stretching efficiency.

in the outlet section, which indicates the high complexity of the multiphase flow in the mixing element.

In summary, the aspect ratio mainly changes the flow distance of the fluid under the same number of elements. As the aspect ratio increases, the shear effect of the fluid in the KSM at the same speed decreases, and then the shear rate decreases, weakening the turbulence of radial secondary flow vortex as well as increasing the distance of fluid re-convergence. The turbulence intensity and energy dissipation rate are reduced, so the SMD of the dispersion phase droplets increases, and the DSD moves to a large size and the distribution is significantly widened. Therefore, increasing the aspect ratio of the mixing element is not conducive to liquid–liquid dispersion and mixing.

4.3.2. Blade twist angle

The effect of blade twist angle on the SMD and DSD is shown in Fig. 10. It can be seen from Fig. 10(a) that the larger the blade twist angle of the mixing element, the smaller the droplet size of the dispersion phase. The dimensionless Sauter mean diameters of the blade twist angles of 90°, 120°, 150° and 180° at the outlet of the KSM are 0.02178, 0.01539, 0.01497 and 0.01410, respectively. From the colormap of the dispersion phase distribution in Fig. 10(a), the droplet size in the inlet section remains basically unchanged. After entering the mixing element section, the droplet is continuously subjected to the strong shear effect of the mixing element and the

turbulent vortex induced by the mixing element, resulting in irregular shear deformation and stretching of the droplet. Meanwhile, the droplet breaks up and becomes smaller. When the average droplet size reaches a stable value, the DSD no longer changes. As the blade twist angle increases, the number of mixing elements required to stabilize the droplet size decreases. With blade twist angles of 90°, 120°, 150° and 180°, the droplet size basically ceased to change after the dispersion phase flowed through the eighth, seventh, sixth, and fourth mixing elements. The turbulent dissipation rate and average energy dissipation rate for blade twist angles of 90°, 120°, 150° and 180° are 2.89 and 13.88 $\text{m}^2 \cdot \text{s}^{-3}$, 4.22 and 19.69 $\text{m}^2 \cdot \text{s}^{-3}$, 5.69 and 25.82 $\text{m}^2 \cdot \text{s}^{-3}$, 7.25 and 33.12 $\text{m}^2 \cdot \text{s}^{-3}$. Combined with Fig. 10(b), the DSD is narrow, and the droplet size is small at a large blade twist angle. With the increase of the blade twist angle, the turbulence intensity increases. The easier the droplet is broken, the smaller the droplet size is, and the energy dissipation rate required increases. The DSD gradually moves towards the smaller size.

The effect of blade twist angle on shear rate and stretching efficiency is shown in Fig. 11. The average shear rates of the blade twist angles of 90°, 120°, 150° and 180° are 892.94, 1008.09, 1123.13 and 1229.31 s^{-1} , respectively, and the average stretching efficiencies are 0.549, 0.557, 0.558 and 0.559, respectively. In Fig. 11(a), the axial distribution of the shear rate of the dispersion phase in KSM is similar to the “U” shape distribution. At the position where

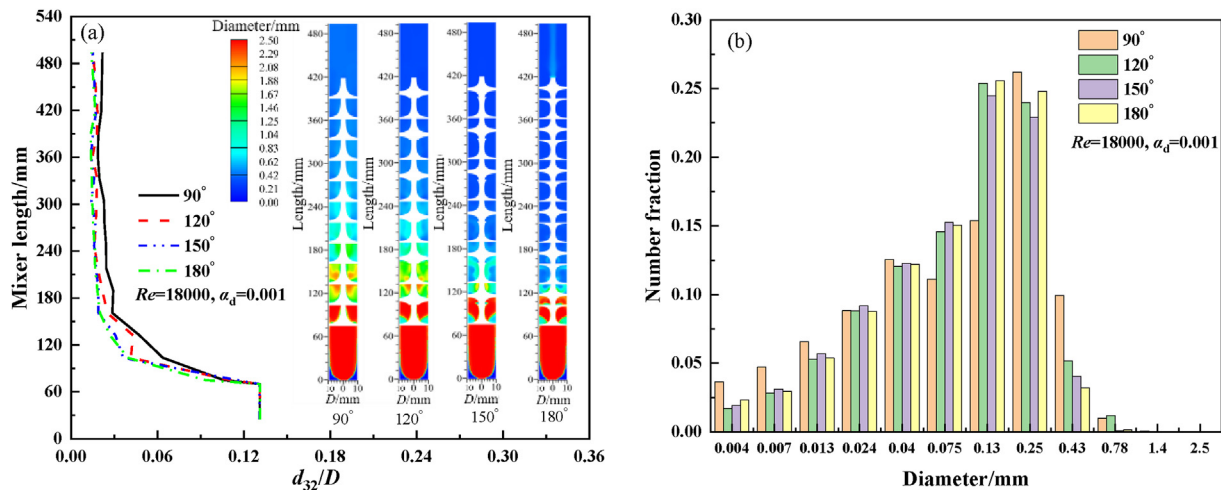


Fig. 10. Effects of blade twist angle on droplet size and its distribution: (a) Sauter mean diameter, (b) droplet size distribution.

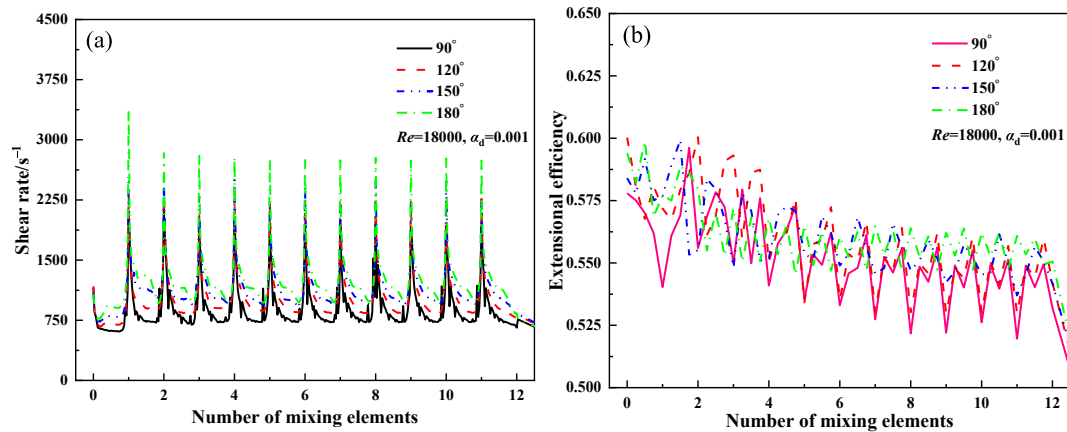


Fig. 11. Effects of blade twist angle on shear rate and stretching efficiency: (a) shear rate, (b) stretching efficiency.

the two mixing elements are at 90° interlaced junction, the twist and deformation of the fluid are caused, thereby increasing the velocity gradient and shear stress of the fluid, and the shear rate peaks, which is conducive to liquid–liquid dispersion and mixing.

As shown in Fig. 11(b), the stretching efficiency shows a sawtooth wave distribution in the KSM. The stretching efficiency in the inlet section (0–75 mm) is higher, and the peak of the mixing element section (75–418.8 mm) is obvious. The stretching efficiency gradually decreases as it flows out of the mixing element section. As the blade twist angle increases, the stretching efficiency increases slightly. The numerical results show that the change of the blade twist angle does not affect the deformation and stretching of the dispersion phase droplets. As the axial flow distance increases, the droplet stretching efficiency gradually decreases.

In summary, the blade twist angle mainly changes the liquid–liquid two-phase flow direction and the turbulence of the fluid in the KSM. With the blade twist angle increased from 90° to 180°, the dispersion phase droplets in the KSM are subject to enhanced shearing action, while the shear rate, turbulence intensity and energy dissipation rate increases. Meanwhile the SMD of the dispersion phase droplets decreases, the DSD moving to a small size and the distribution becoming narrower. Therefore, increasing the blade twist angle of the mixing element is beneficial to the liquid–liquid dispersion and mixing.

4.3.3. Blade assembly angle

The effect of blade assembly angle on Sauter mean diameter and droplet size is shown in Fig. 12. As it can be seen in Fig. 12(a), the

SMD decreases as the blade assembly angle increases. The dimensionless Sauter mean diameters at the outlet of KSM with blade assembly angles of 0°, 45° and 90° are 0.01943, 0.01758 and 0.01497, respectively. The dispersion phase swirls at the outlet of the mixing element section relying on inertia force, and the velocity decreases at the center of the pipe. Currently, the breakage rate and the energy dissipation rate decreases, and the droplet size increases. It can be seen from the colormap of the dispersion phase distribution in Fig. 12(a), as the axial flow distance increases, the difference in DSD gradually decreases. As the blade assembly angle increases, the number of mixing elements required to stabilize the droplet size decreases. When the blade assembly angles are 0°, 45° and 90°, the droplet size is basically unchanged after the dispersion phase flows through the 5th, 4th, and 3rd mixing elements, it is shown in Fig. 12(b) that the DSD is narrow, and the droplet size is small under the larger blade assembly angle. With the increase of the blade assembly angle, the droplet size is smaller, and the DSD gradually shifts towards the smaller size.

The effect of blade assembly angle on shear rate and stretching efficiency is shown in Fig. 13. The average shear rates of the blade assembly angle of 0°, 45° and 90° are 878.75, 1135.59 and 1229.31 s^{-1} , respectively, and the average stretching efficiencies are 0.565, 0.559 and 0.558, respectively. It can be seen in Fig. 13(a) that for the mixing elements are at 0° interlaced junction, the fluctuation of shear rate at different positions is not large and basically close. As the blade assembly angle increases, the overall shear rate in the KSM, the shear effect of the fluid and the peak shear rate at the contact position of the mixing element increases. In Fig. 13(b),

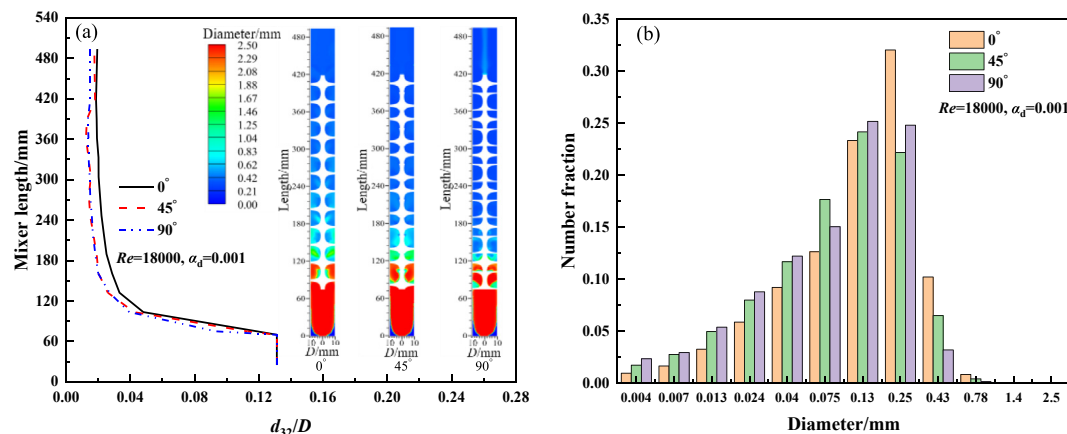


Fig. 12. Effects of blade assembly angle on droplet size and its distribution: (a) Sauter mean diameter, (b) droplet size distribution.

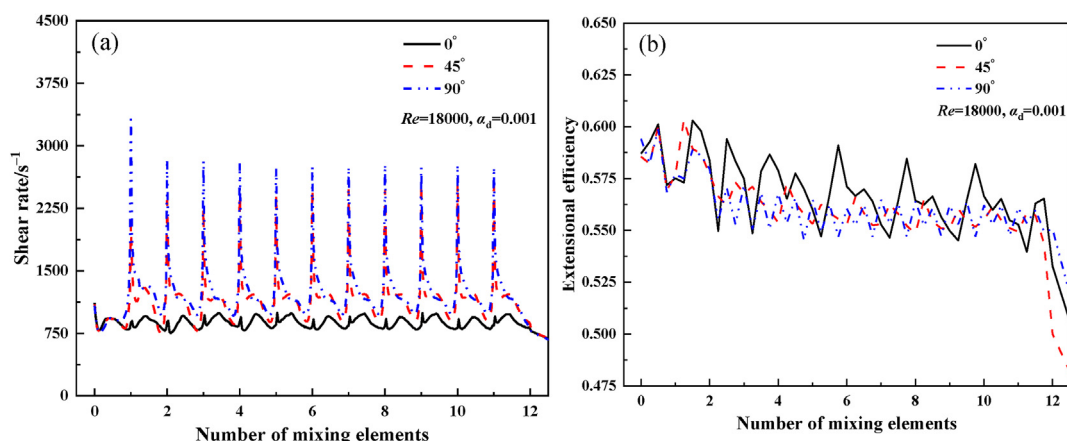


Fig. 13. Effects of blade assembly angle on shear rate and stretching efficiency: (a) shear rate, (b) stretching efficiency.

the stretching efficiency in the KSM shows a sawtooth wave trend. The stretching efficiency is larger in the inlet section and upstream of the mixing element section, and relatively small in the outlet section. The stretching efficiency of the assembly angle of 0° and 45° decreases rapidly in the outlet section.

In summary, as the blade assembly angle increases, the shear effect of the fluid in the KSM increases and the shear rate increases as well, which enhances the turbulence of the fluid and decreases the SMD of the dispersion phase droplets. The DSD gradually shifts towards the smaller size and the distribution is significantly narrowed. Therefore, increasing the blade assembly angle of the mixing element is beneficial to liquid–liquid dispersion and mixing.

4.3.4. Swirl direction of blade

The effect of the swirl direction of the blade on the SMD and droplet size is shown in Fig. 14. In Fig. 14(a), the dimensionless Sauter mean diameters at the outlet of KSM are 0.01497, 0.05672 and 0.08760, respectively, when the swirl direction of the blade is different spin of 90° , same spin of 90° and same spin of 0° . Compared with the mixing elements with the same spin, the mixing elements with different spin show different two-phase fluid behaviors. As the axial flow distance of the two-phase fluid increases, the dispersion phase is dispersed in the continuous phase with smaller droplets, which is more conducive to liquid–liquid dispersion and mixing. This is similar to the research results of Hobbs and Muzzio [34,35]. When the swirl direction of the mixing element is the same, the liquid–liquid two-phase dispersion effect

with a blade assembly angle of 90° is better than 0° . It can be seen from the colormap distribution of the dispersion phase in Fig. 14(a) that for the mixing elements of different spin of 90° and same spin of 90° , the droplet size gradually stabilizes when the dispersion phase flows through the 3rd and 10th mixing elements. For the mixing element with same spin of 0° , the dispersion phase is not sheared by the element. The dispersion phase droplets break up under the effect of the turbulent vortex, and the droplet breakage frequency is reduced, which is not conducive to the dispersion and mixing of droplets. Fig. 14(b) shows that compared with the same spin, the dispersion phase droplets of the mixing element in the different spin are concentrated in smaller droplets, and as the blade assembly angle increases, the droplet size becomes smaller, and the DSD gradually shifts towards the smaller size.

The effect of swirl direction on shear rate and stretching efficiency is shown in Fig. 15. It can be seen from Fig. 15 that the average shear rates of different spin of 90° , same spin of 90° and same spin of 0° are 1229.31, 938.72 and 521.65 s^{-1} , respectively, and the average stretching rate are 0.558, 0.522 and 0.516, respectively. As shown in Fig. 15(a), the shear rate of the mixing elements with different spin of 90° is larger than that of the mixing elements with same spin of 90° and is much larger than that of the mixing elements with same spin of 0° . Because the mixing elements with different spin of 90° , same spin of 90° and same spin of 0° have radial spin, the mixing elements with different spin of 90° and same spin of 90° have a peak shear rate at the 90° interlaced junction of the two mixing elements. However, the mixing

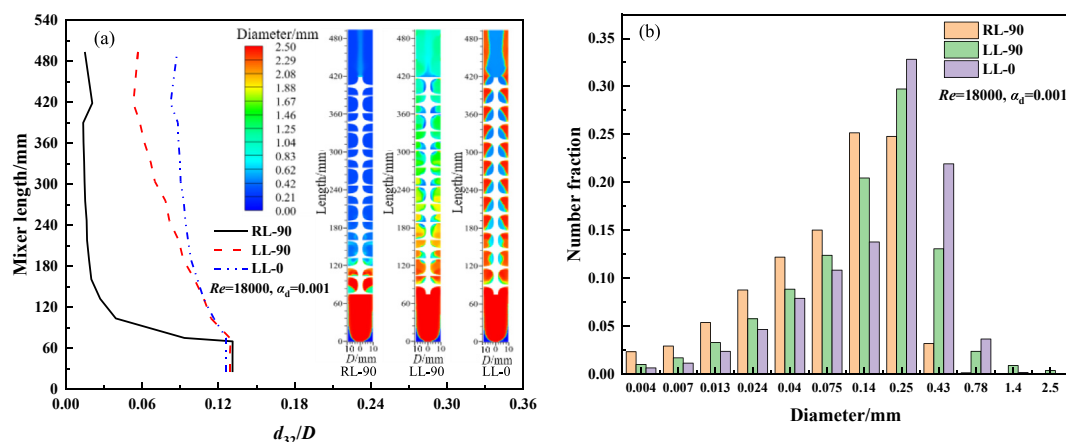


Fig. 14. Effects of blade swirl direction on droplet size and its distribution (R: Right-handed, L: Left-handed): (a) Sauter mean diameter, (b) droplet size distribution.

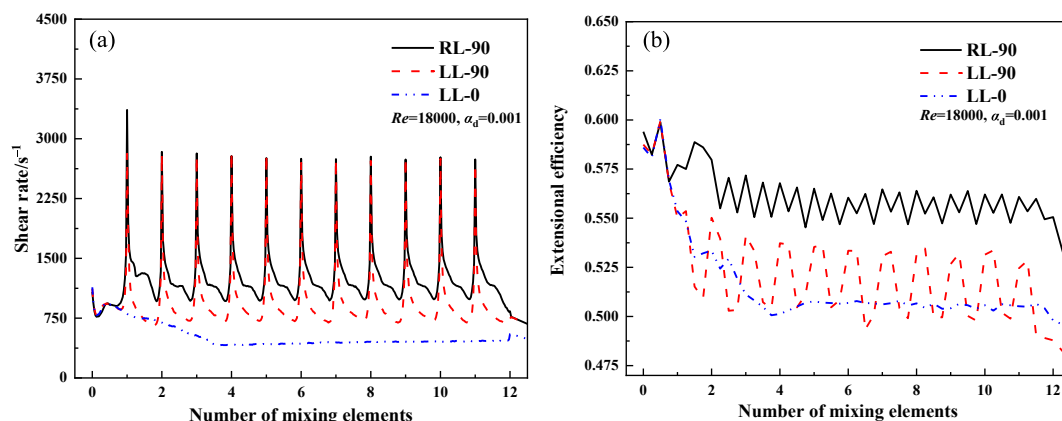


Fig. 15. Effects of blade swirl direction on shear rate and stretching efficiency: (a) shear rate, (b) stretching efficiency.

Table 3
Comparison of mixing elements structures

Aspect ratio	Blade twist angle/(°)	Blade assembly angle/(°)	Swirl direction	SMD/mm	$\Delta P/\text{Pa}$
0.5	180°	90°	Different spin	0.0060	160651
1.0	180°	90°	Different spin	0.0106	44024
1.5	180°	90°	Different spin	0.0140	22705

elements with same spin of 0° cannot shear the shunt and change the spin flow direction, so its shear rate is much smaller than the other two, and there is no peak. With the increase of the axial distance, the shear rate gradually decreases, and does not change significantly after flowing through the 4th mixing element.

It can be seen from Fig. 15(b) that the stretching efficiency of the mixing elements with different spin of 90° are the largest, that of the mixing elements with same spin of 90° are the second, and that of the mixing elements with same spin of 0° are the smallest. The axial arrangement of the three mixing elements has a strong deformation and stretching effect on the dispersion phase droplets upstream of the mixing element section (0–4th mixing elements). This is because the dispersion phase droplets are obviously broken in the upstream of the mixing element section. With the increase of the axial position, the dispersed droplet size gradually stabilizes. In addition, since the mixing elements with same spin of 0° only have the effect of radial spin, the flow field distribution is more uniform and the shear stress is smaller, which is not conducive to the deformation and stretching of the droplets.

In summary, the different spin mixing elements have higher shear rate and stretching efficiency, and the fluid turbulence is stronger. The SMD is 3.7–5.8 times smaller than that in the mixing elements with same spin. The DSD is concentrated in small size and the distribution is obviously narrow. Therefore, the use of the mixing elements with different spin can significantly improve the liquid–liquid dispersion and mixing.

4.3.5. Structure optimization

From the above research results, it is concluded that the aspect ratio has the most significant effect on the SMD and DSD of the dispersion phase droplets. Therefore, the aspect ratio is an important structural parameter in the optimization process which needs to be focused on. It can be seen from Table 3 that when the aspect ratio is 0.5, the SMD of the dispersion phase droplets is 1.76–2.33 times smaller than that corresponding to other aspect ratios, and the DSD is narrower, but the pressure drop of KSM is 3.65–7.07 times larger than that of other aspect ratios. Therefore, at excessive small values of the aspect ratio, a notable escalation in pressure

drop occurs, followed by the increase of energy consumption. Taking into account the pressure drop and energy consumption in a comprehensive manner, the KSM with an aspect ratio of 1–1.5 is more suitable for actual production needs. In summary, the structural parameters of the optimized mixing elements are as follows: aspect ratio of 1–1.5, blade twist angle of 180°, blade assembly angle of 90°, and interlaced assembly of adjacent elements with different spin.

5. Conclusions

Based on the CFD–PBM coupled method, this work studies the effects of operating conditions and structural parameters of KSM on DSD and droplet size, to further reveal the liquid–liquid dispersion characteristics. Combined with the efficiency and accuracy of the simulation calculation, 12 Bins with a droplet size range of 0.004–2.506 mm was determined as the discrete size range of the dispersion phase. The simulation results show that as the Reynolds number increases, the energy dissipation rate will increase, and the dispersion phase droplets are easily broken and become smaller. The DSD shifts towards the smaller size. With the increase of the volume fraction of the dispersion phase, the probability of droplet collision increases, and the droplets are easy to coalesce and become bigger. The DSD curve shifts towards the larger size. With the decrease of the aspect ratio of the mixing element, the shear effect of the fluid in the KSM becomes stronger. The shear rate, the turbulence intensity and the energy dissipation rate increase, the SMD of the dispersion phase droplets decreasing, and the DSD shifts towards the smaller size and the distribution is obviously narrowed. As the blade twist angle increases, the shear rate, the turbulence intensity and the energy dissipation rate increase, and then the SMD of the dispersion phase droplets decreases. The DSD shifts towards the smaller size and the distribution becomes narrower. As the blade assembly angle increases, the shear rate in the KSM increases, which enhances the turbulence of the fluid, resulting in a decrease in the SMD of the dispersion phase droplets and the DSD shifts towards the smaller size and the distribution becomes narrower. It is worth noting that the different spin mixing

element has higher shear rate and stretching efficiency, and the fluid turbulence is stronger. The SMD is 3.7–5.8 times smaller than that of the same spin mixing element, and the DSD is concentrated on small sizes. The distribution is significantly narrower. Taking into account the pressure drop and efficiency in a comprehensive manner, the structural parameters of the optimized mixing elements are as follows: aspect ratio of 1–1.5, blade twist angle of 180°, blade assembly angle of 90°, and interlaced assembly of adjacent elements with different spin. This work provides an important theoretical support and reference for future structural optimization and process enhancement of KSM.

Declaration of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT Authorship Contribution Statement

Junhai Deng: Writing – original draft, Writing – review & editing. Shilin Lan: Investigation. Juchang Wu: Software, Validation, Writing – original draft. Shenghua Du: Methodology. Weidong Liu: Supervision. Luchang Han: Methodology, Resources. Yefeng Zhou: Supervision, Funding acquisition.

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Supplementary Material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cjche.2024.03.018>.

Nomenclature

a	interfacial area, $\text{m}^2 \cdot \text{m}^{-3}$
$a(d_i, d_j)$	collision rate, $\text{m}^3 \cdot \text{s}^{-1}$
D	tube diameter, m
d	droplet diameter, m
d_{32}	Sauter mean diameter, m
g	acceleration of gravity, $\text{m} \cdot \text{s}^{-2}$
$h(d, d')$	aggregation kernel, $\text{m}^3 \cdot \text{s}^{-1}$
k	turbulent kinetic energy, $\text{m}^2 \cdot \text{s}^{-2}$
L	tube length, m
n_i	number of droplets
Re	Reynolds numbers
S	source term of equation
u	velocity, $\text{m} \cdot \text{s}^{-1}$
α	phase volume fraction
γ	shear rate, s^{-1}
Γ	breakage frequency, s^{-1}
ε	turbulent dissipation rate, $\text{m}^2 \cdot \text{s}^{-3}$
η	Kolmogorov scale
μ	molecular viscosity, $\text{Pa} \cdot \text{s}$
v_s	slip velocity, $\text{m} \cdot \text{s}^{-1}$
ρ	density, $\text{kg} \cdot \text{m}^{-3}$

σ	interfacial tension, $\text{N} \cdot \text{m}^{-1}$
ω	specific dissipation rate, s^{-1}
Ω	breakage rate, $\text{m}^{-3} \cdot \text{s}^{-1}$

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